

Oluwajolasun Jaiyesimi

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SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, React, FastAPI, Django, Node.js, SQL (PostgreSQL), NoSQL (Elasticsearch).
Tools & Platforms: GCP, AWS, Docker, Git, GitHub, GitHub Actions, Honeycomb, Pub/Sub, Postman.
Technical & Soft Skills: Full-Stack Development, REST APIs, Webhooks, Event-Driven Architectures, ETL Pipelines, Microservices, Machine Learning (TensorFlow, Keras, Hugging Face), Testing (Pytest, Unit/Integration/E2E), CI/CD, Agile/Scrum, Software Development Lifecycle, Design Patterns, Data Visualization, Spatial Data Analysis, Cross-functional Collaboration.

WORK EXPERIENCE

Areto Labs

Edmonton, Canada.

Software Engineer - Full Stack Web Developer

July 2025 – Present

- Designed and deployed a GCP microservice using a fine-tuned vision model, increasing spam-bot detection accuracy by 60% and improving comment moderation efficiency across platforms.
- Trained and fine-tuned a custom Hugging Face model, improving classification accuracy and reducing false positives, resulting in fewer manual reviews and faster moderation throughput
- Architected event-driven data ingestion systems using Google Pub/Sub to stream topic and subscription messages between microservices, leveraging Honeycomb traces to monitor, debug, and optimize event flows.
- Implemented and maintained DevOps pipelines using GCP, GitHub, and GitHub Actions, improving deployment speed and reliability.
- Led code reviews and contributed to architectural decisions that improved reliability and reduced production defects.
- Optimized React and TypeScript front-end applications to deliver seamless, reduced load time by ~20%.

CANN Forecast

Montreal, Quebec.

Software Engineer - Full Stack Web Developer

September 2023 – September 2024

- Built ETL pipelines and performed geospatial analysis for 10+ municipal water infrastructure projects, enabling planners to make data-driven operational decisions.
- Collaborated with an agile team of domain experts to develop React based interfaces, interactive maps using Leaflet.js and Mapbox, and visualizations with D3.js, used by 100+ stakeholders, significantly improving accessibility and decision-making speed.
- Automated email attachment processing via Python (IMAP/SMTP), cutting manual work by 90%, and integrated the solution with Jenkins, resulting in faster deployment cycles and reduced processing time by 4 hours per day.
- Engineered scalable backend systems in Django, exposing RESTful APIs that supported dynamic front-end functionality, serving over 500 concurrent users.

Hifi Engineering

Calgary, Alberta.

Software Engineer - Machine Learning Engineer

September 2022 – July 2023

- Designed and implemented ML models for sensor data analysis, achieving 95% accuracy in simulated pipeline leak detection using Keras and TensorFlow.
- Reduced false positives by 15% through close collaboration with domain experts during model tuning and optimization.
- Improved system sensitivity by 20% via rigorous validation, significantly enhancing real-world reliability.
- Communicated technical results to both technical and non-technical stakeholders, contributing to a projected 30% reduction in operational risk.

EDUCATION

Master's Degree, Software Engineering (MEng)

May 2021 – June 2023

University of Calgary

Calgary, Alberta.

- GPA: 3.85 / 4.00

Bachelor's Degree, Electrical and Electronics Engineering (BEng)

October 2015 – July 2018

Loughborough University

Loughborough, United Kingdom.

- 2nd Class Upper (GPA Equivalent: 3.72 / 4.00)

TECHNICAL PROJECTS

AI-Powered Content Moderation Microservice: (Private Repository)

- Built a FastAPI microservice using a fine-tuned Hugging Face model to automatically detect unsafe profile images and filter spam.
- Integrated Apify scraping, Pub/Sub messaging, and OpenTelemetry/Honeycomb tracing for a scalable, production-ready workflow.

Novel CNN for Fish Classification (FishNet): https://github.com/oluwajolasun/ENEL645_FinalProject_FishClassification

- Designed a custom CNN ("Fishnet") achieving 99.3% test accuracy with regularization techniques, to reduce overfitting and improve generalization.
- Benchmarked against VGG16 and EfficientNetB1 using transfer learning on a 9,430-image dataset.